

Transformer vs. CNN: Hyperparameter-Tuned Deep Learning Models for Automated Malaria Diagnosis

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Abstract

Malaria remains a pressing global health issue, particularly in sub-Saharan Africa, where it accounts for approximately 95% of malaria-related deaths. Early and accurate diagnosis is critical but often limited in low-resource settings due to a lack of trained personnel and diagnostic infrastructure. This study investigates the application of deep learning models for the automated detection of malaria in microscopic blood smear images. We compare two state-of-the-art architectures: ResNet50, a well-established convolutional neural network, and the Swin Transformer, a hierarchical vision transformer. Both models were pre-trained on ImageNet and evaluated using 5-fold cross-validation on a balanced, expert-annotated public dataset. Performance metrics—including accuracy, precision, recall, F1 score, and loss—were analyzed, with a particular focus on recall stability to ensure reliable detection of infected samples. Our results show that while both models performed well, the Swin Transformer consistently outperformed ResNet50 in terms of recall consistency and convergence. Further optimization using the Optuna framework significantly enhanced the Swin Transformer's performance, reducing false positives and false negatives and improving training stability. These findings highlight the potential of transformer-based architectures for high-accuracy, scalable malaria diagnosis.

Keywords

deep learning, malaria, transformers, convolution neural network

1 Introduction

Malaria continues to pose one of the most pressing global public health threats, particularly in regions with limited access to healthcare infrastructure and preventive measures [11]. It is a potentially life-threatening disease caused by protozoan parasites of the genus *Plasmodium*, transmitted to humans primarily through the bite of *Anopheles mosquitoes* [5]. Among the five *Plasmodium* species known to infect humans, *P. falciparum*, *P. vivax*, and *P. malariae* are the most prevalent, with *P. falciparum* being the deadliest, responsible for the majority of severe cases and malaria-related deaths.

The initial symptoms of malaria are typically non-specific, including fever, chills, sweating, headaches, and myalgia. As the disease progresses, it can present with gastrointestinal symptoms, fatigue, and potentially life-threatening organ complications [4]. According

to the World Malaria Report 2024 by the World Health Organization (WHO) [15], malaria caused an estimated 263 million cases and 597,000 deaths globally in 2024. Alarming, 95% of these fatalities occurred in Africa, predominantly among populations with poor access to diagnostic and treatment services.

Various diagnostic methods are currently used to detect malaria [3, 10]. Among them, polymerase chain reaction (PCR) offers high sensitivity and specificity by amplifying *Plasmodium* DNA [8], but its cost and technical demands make it impractical for widespread use in low-resource settings. Microscopic examination of stained blood smears remains the gold standard in many areas due to its accessibility and low cost. In this method, parasites are visually identified within red blood cells.

Despite its advantages, microscopy has significant drawbacks: it is labor-intensive, dependent on trained personnel, and subject to inter-observer variability and human error, especially in high-volume diagnostic settings. These limitations highlight the need for automated diagnostic solutions that are both efficient and scalable. In this context, deep learning has emerged as a powerful tool to enhance image-based diagnosis by enabling fast, accurate, and reproducible analysis of blood smear images [7].

Historically, Convolutional Neural Networks (CNNs) have been the backbone of medical image classification due to their strong performance in spatial feature extraction. Recently, Vision Transformers (ViTs) have gained attention for their ability to model long-range dependencies in image data using self-attention mechanisms [13]. In this work, we conduct a comparative study between a classical CNN model (ResNet50) [16] and a modern Vision Transformer (Swin Transformer) for the task of malaria detection in microscopic blood cell images. Our goal is to assess their diagnostic performance and explore the advantages and trade-offs of these two deep learning approaches in a critical real-world medical application.

2 Methodology

In this study, we investigate the performance of two distinct deep learning architectures for classifying red blood cell images as parasitized or uninfected using a publicly available malaria dataset. The models evaluated are the Swin Transformer, a state-of-the-art vision transformer, and ResNet50, a widely used convolutional neural network (CNN).

2.1 Swin Transformer

We implement the Swin Transformer (Shifted Window Transformer), a hierarchical vision transformer architecture designed for computer vision tasks [9]. Unlike conventional transformers that apply global self-attention—resulting in high computational cost for image data—the Swin Transformer introduces a shifted window attention mechanism. It partitions the image into non-overlapping windows, performs self-attention within each window, and shifts

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the windows in subsequent layers to capture cross-window dependencies. This strategy reduces computational complexity while effectively modeling both local and global contextual features, making it well-suited for image classification and related tasks [14].

2.2 ResNet50

We also employ ResNet50, a 50-layer convolutional neural network that introduces residual connections to address the degradation problem in deep learning [6]. These shortcut connections allow gradients to flow more efficiently through the network, facilitating the training of deeper architectures and enabling the model to learn more complex representations. ResNet50 is widely recognized for its strong performance in computer vision and serves as a robust CNN baseline in our experiments.

2.3 Transfer Learning and Hyperparameter Optimization

To maximize performance and training efficiency, both models are initialized using pre-trained weights from the ImageNet dataset, leveraging transfer learning to reuse general-purpose visual features. This approach reduces training time, mitigates overfitting, and enhances the models' ability to generalize on the malaria dataset.

For further refinement, we apply hyperparameter optimization using Optuna [1], a state-of-the-art optimization framework that employs Bayesian methods to navigate the hyperparameter space efficiently. For ResNet50, we tune parameters including learning rate, optimizer type, and batch size. For the Swin Transformer, we adjust training-specific parameters to better accommodate its hierarchical and windowed attention mechanisms.

This combination of transfer learning and automated hyperparameter search enhances model convergence, training stability, and classification accuracy—critical for reliable diagnostic performance in a high-stakes medical context.

3 Experiments and Results

In this study, we evaluated the performance of two deep learning models on the task of binary classification of malaria-infected blood cells: a convolutional ResNet model and a vision Transformer model (Swin Transformer), both pre-trained on ImageNet. All experiments were performed using the PyTorch framework.

3.1 Dataset Description

The dataset used in this study is the Malaria Cell Images Dataset provided by the National Library of Medicine (NLM) [12]. It contains 27,558 labeled microscopic images of blood smears, evenly split between parasitized (13,779) and uninfected (13,779) samples. The dataset includes infections from *Plasmodium vivax* and *Plasmodium falciparum*. Images were captured using a smartphone-based imaging system developed by the NLM, in which an Android device, using the Malaria Screener app, records images directly from a microscope's field of view. All images were manually annotated by domain experts, ensuring label accuracy and consistency. The dataset is balanced, publicly available, and intended for use in malaria diagnosis research.

Although the dataset is balanced, to further mitigate the risk of overfitting and ensure robust model evaluation, we employed the K-Fold Cross-Validation technique [2]. The data was divided into k equally sized folds. The model was trained k times, each time using $k - 1$ folds for training and one fold for validation. This approach ensures that every data point is used for both training and validation, providing a more reliable estimate of model generalization. For this study we considered $k=5$.

3.2 Training

We recorded key performance metrics per epoch and fold, including accuracy, precision, recall, F1-score, loss, and the confusion matrix components. Particular emphasis was placed on the recall metric, as it reflects the model's ability to detect true positive malaria cases—a crucial aspect in medical diagnosis.

As shown in Figure 1, epoch 20 appeared to be the most stable point across folds, with recall values for both models converging between 0.97 and 0.98. The Swin Transformer's Fold 2 demonstrated peak performance (>0.98) between epochs 10 and 15, while ResNet50's Fold 5 showed high and stable recall throughout training.

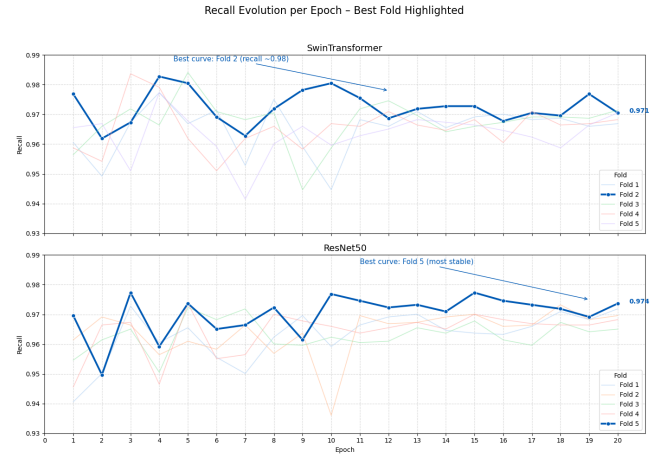


Figure 1: Evolution of recall per epoch for Swin Transformer and ResNet50 across 5-fold cross-validation. Each line represents one fold. Best folds: Fold 2 (Swin) and Fold 5 (ResNet).

To further assess generalization, we selected the best-performing folds from each model (Fold 2 for Swin, Fold 5 for ResNet) and evaluated them against a held-out test set (Fold 6). Figure 2 presents the corresponding confusion matrices. Both models showed excellent performance. While the Swin Transformer had fewer false negatives during training, ResNet50 achieved slightly fewer false positives on the test set (67 vs. 84), highlighting its more conservative predictions for the negative class. Importantly, both models had the same number of false negatives (86) in the test, showing balanced sensitivity to malaria-positive cases.

Given its strong baseline, we selected the Swin Transformer for further optimization using Optuna.



Figure 2: Confusion matrices for Swin Transformer (top) and ResNet50 (bottom). Left: Training performance on best fold. Right: Test performance on Fold 6.

3.3 Hyperparameter Optimization with Optuna

We applied the Optuna framework to optimize the Swin Transformer by maximizing the recall score. The search space included:

- Learning rate: $[1e-6, 1e-3]$;
- Weight decay: $[1e-6, 1e-2]$;
- Loss function: CrossEntropyLoss, BCELoss;
- Optimizer: Adam, RMSprop, SGD;

A total of 30 trials were conducted. Figure 3 compares the recall evolution between the baseline and the optimized model. The Optuna-optimized Swin Transformer achieved a mean recall approx. 0.991, with significantly smoother curves, indicating enhanced training stability and generalization.

Confusion matrices for both models on Fold 2 are shown in Figure 4. The Optuna-optimized model demonstrated lower error rates in both classes, with improved precision and recall, underscoring the value of hyperparameter tuning.

We also analyzed the loss curves for Fold 2 (Figure 5). The Optuna-tuned model converged faster and exhibited smoother, lower loss across epochs, indicating more efficient and stable training.

Table 1 presents a summary of performance metrics for the best folds of each model, including training and test results. The Swin Transformer with Optuna achieved the highest scores across all metrics, especially recall and F1-score, confirming its superior performance.

All experiments were conducted on a workstation running Windows 11 Home, equipped with an AMD Ryzen 9 7900X CPU, 64 GB of RAM, and an NVIDIA GeForce RTX 4070 Ti Super GPU (16 GB VRAM). This setup provided sufficient computational power for both training and evaluation tasks.

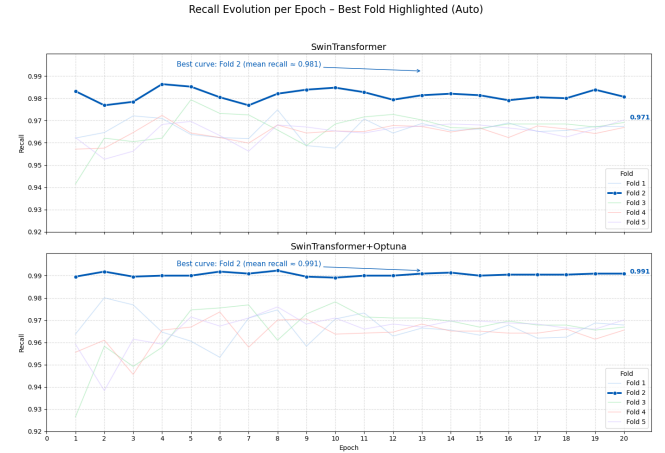


Figure 3: Recall per epoch for Swin Transformer with and without Optuna. Each curve represents one fold. Optuna improves recall and stability, especially in Fold 2.

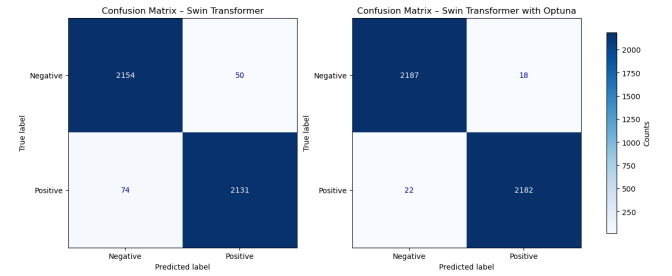


Figure 4: Confusion matrices of Swin Transformer without (top) and with (bottom) Optuna optimization on Fold 2. Optuna reduces both false positives and false negatives.

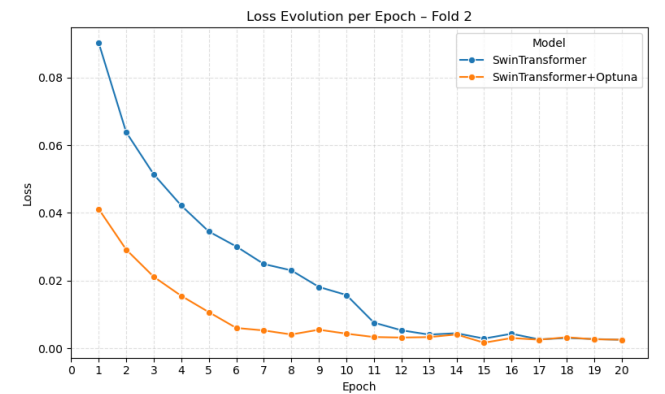


Figure 5: Training loss per epoch for Swin Transformer with and without Optuna (Fold 2). Optuna results in faster convergence and lower final loss.

Table 1: Performance summary for Swin Transformer and ResNet50 (best folds only).

Model	Accuracy	Precision	Recall	F1-score	Loss
ResNet50 – Train (Fold 5)	0.9751	0.9785	0.9696	0.9740	0.00014
ResNet50 – Test (Fold 6)	0.9764	0.9755	0.9688	0.9721	0.17229
Swin Transformer – Train (Fold 2)	0.9740	0.9616	0.9769	0.9692	0.00251
Swin Transformer – Test (Fold 6)	0.9745	0.9695	0.9688	0.9691	0.25534
Swin Transformer with Optuna – Train (Fold 2)	0.9927	0.9936	0.9918	0.9927	0.00603

4 Conclusions

This study presented a comparative evaluation of two deep learning architectures—ResNet50, a classical convolutional neural network, and the Swin Transformer, a modern vision transformer—for the automated diagnosis of malaria from microscopic blood smear images. Both models demonstrated strong classification performance, but the Swin Transformer consistently outperformed ResNet50 in terms of recall stability and loss convergence, particularly across multiple cross-validation folds. These findings motivated the selection of the Swin Transformer for further refinement through hyperparameter optimization.

By applying Optuna for automated hyperparameter tuning, we achieved notable improvements in the Swin Transformer's performance. The optimized model showed faster convergence, lower loss, and more stable recall over training epochs. It also achieved fewer false positives and false negatives, as evidenced by confusion matrix analysis, thus reinforcing its potential for reliable application in clinical diagnostics.

These results highlight the Swin Transformer's promise as an effective and scalable solution for image-based malaria detection. As future work, we intend to evaluate the model's robustness on imbalanced datasets, better simulating real-world epidemiological distributions. We also plan to explore its extension to segmentation tasks in whole-slide microscopic images, contributing to the development of end-to-end diagnostic pipelines for malaria and potentially other infectious diseases.

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